

CIND820 Milestone2Interim Report

Data Profiling and Audit

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1. Project overview

1.1 Dataset Overview

The project uses the Toronto Neighbourhood Profiles dataset, which contains socioeconomic indicators for 140 Toronto neighbourhoods. The dataset consists of structured tabular data where each row represents a neighbourhood and each column represents a socioeconomic indicator. The selected indicators include measures of income, employment, education, housing affordability, immigration, visible minority populations, and family structure.

1.2 Data Structure

The dataset is structured data in CSV format. No semi-structured or unstructured data sources were used in this phase of the project. The dataset was imported into Python using Pandas and transformed into a machine-learning-ready format for subsequent analysis.

1.3 Univariate Analysis

Histograms were generated for all selected indicators to examine their distributions. Several indicators exhibited positive skewness, particularly housing and demographic variables such as Visible Minority population, Immigrants, Renters, and Lone Parent Families. Employment-related variables displayed more symmetric distributions. The variability observed across indicators justified the need for standardization prior to dimensionality reduction.

1.4 Bivariate Analysis

Correlation analysis revealed several strong relationships among the selected indicators. Immigrants and Visible Minority populations showed a very strong positive correlation ($r = 0.97$). Participation Rate and Employment Rate were highly positively correlated ($r = 0.99$). Employment Rate and Unemployment Rate showed a strong negative relationship ($r = -0.77$). These findings indicate substantial overlap among several indicators and suggest the presence of multicollinearity.

1.5 Constraints and Limitations

Variance Inflation Factor (VIF) analysis identified severe multicollinearity among multiple indicators. Participation Rate and Employment Rate exhibited extremely large VIF values, indicating redundancy in the information captured by these variables. As a result, Principal Component Analysis (PCA) was selected to reduce dimensionality and mitigate multicollinearity prior to clustering analysis. Another limitation is that the dataset represents a single point in time and therefore cannot support forecasting or causal claims. The results should be interpreted as descriptive patterns rather than evidence of causation.

1.6 Privacy and Ethics Review

The dataset contains aggregated neighbourhood-level information and does not include any personally identifiable information (PII). No individual residents can be identified from the dataset. The analysis therefore presents minimal privacy risk and aligns with the Tri-Council Policy Statement (TCPS 2) guidance regarding the use of publicly available aggregated data.

2. Methodological Rationale

The objective of this project is to identify socioeconomic factors associated with low-income prevalence and to classify Toronto neighbourhoods according to their socioeconomic vulnerability. For this, the following methods were chosen- a combination of Exploratory Data Analysis (EDA), correlation analysis, Principal Component Analysis (PCA), and K-Means clustering. The main reason is they provide transparent, interpretable, and reproducible results while satisfying the project's requirement to avoid "black box" approaches.

A common approach in this socioeconomic research is the use of simple correlation analysis and descriptive statistics to identify relationships between variables. Correlation analysis is useful for identifying linear relationships between indicators such as unemployment rates, housing affordability, educational attainment, and low-income prevalence. But this correlation analysis alone has important limitations. Socioeconomic indicators are often highly correlated with one another, resulting in multicollinearity. For example, neighbourhoods with higher unemployment rates may also exhibit lower educational attainment and higher housing affordability challenges. Examining variables individually may therefore produce redundant or misleading interpretations.

So in order to address these limitations, Principal Component Analysis (PCA) was selected as a dimensionality reduction technique. PCA transforms multiple correlated indicators into a smaller number of orthogonal principal components while retaining most of the variation present in the original dataset. According to Jolliffe and Cadima (2016), PCA is one of the most widely used techniques for reducing dimensionality and identifying underlying patterns within multivariate datasets. So in our analysis, by reducing the original 12 socioeconomic indicators into 3 principal components explaining approximately 89% of the total variance, the analysis becomes more interpretable while minimizing information loss.

- 1) First principal component (PC1) was found to represent socioeconomic vulnerability, combining high levels of immigrants, visible minority populations, housing need, unemployment, and lone-parent families.
- 2) The second principal component (PC2) captured educational attainment and labour market participation.
- 3) The third principal component (PC3) reflected housing affordability pressures and low-income prevalence.

These components provide a more concise representation of neighbourhood socioeconomic conditions than the original indicators.

While PCA summarizes patterns within the data, it does not directly classify neighbourhoods into meaningful groups. Therefore, K-Means clustering was selected as the primary segmentation method. K-Means is a transparent and widely accepted unsupervised learning algorithm that groups observations according to similarity in their characteristics. Unlike more complex clustering approaches, K-Means produces easily interpretable cluster assignments and can be readily explained to policy makers and stakeholders.

Also, other alternative clustering methods were considered. Hierarchical clustering provides a visual representation of relationships between observations but becomes increasingly difficult to interpret with larger datasets. Density-based approaches such as DBSCAN can identify irregularly shaped clusters but require parameter tuning and may produce unstable results when applied to socioeconomic data with varying densities. Given the relatively small sample size of 140 Toronto neighbourhoods and the project's emphasis on interpretability, K-Means was determined to be the most appropriate clustering technique.

The selected methodology also aligns with established approaches used in vulnerability and deprivation studies. Previous research has frequently combined PCA with clustering techniques to construct socioeconomic vulnerability indices and identify communities with similar social and economic characteristics. This combination enables researchers to reduce redundancy among indicators while simultaneously identifying meaningful population segments for policy analysis.

While the current analysis focuses on cross-sectional neighbourhood characteristics, future project phases will extend the framework using historical neighbourhood data from 2006 to support temporal analysis and predictive modelling.

Another important consideration was methodological transparency. The project follows the "no black box" principle outlined in the course requirements. Every stage of the analytical process can be explained and reproduced, from data cleaning and standardization to principal component extraction and cluster assignment. The retained principal components can be interpreted through their loadings, and cluster characteristics can be understood by examining average indicator values within each group. This transparency is particularly important when results may inform public policy, resource allocation, or community intervention strategies.

Overall, the selected methodology provides a balanced approach that combines statistical rigor with interpretability. EDA and correlation analysis establish an understanding of the data, PCA reduces dimensionality while preserving variance, and K-Means clustering identifies neighbourhood groups with distinct socioeconomic profiles. Compared with a baseline correlation-only approach, the chosen framework captures more complex relationships among indicators and produces actionable insights regarding neighbourhood vulnerability across Toronto.

Figure 1. Correlation matrix showing relationships among selected socioeconomic indicators.

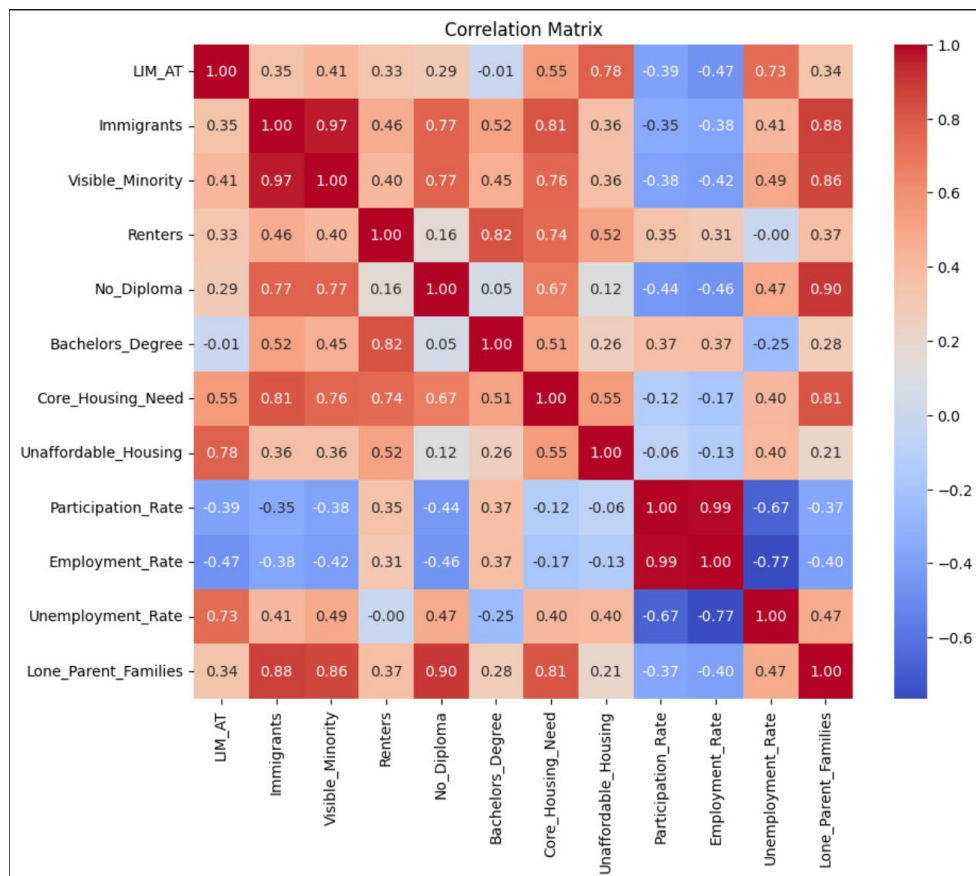
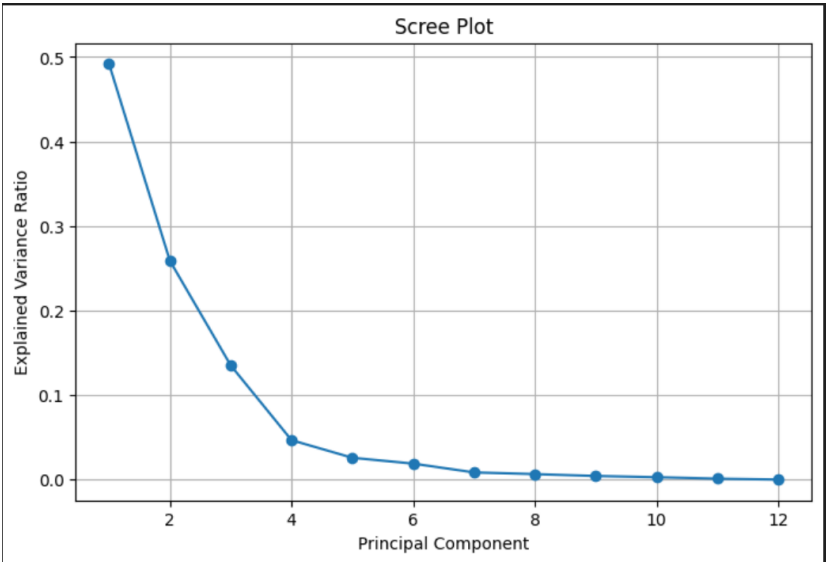


Figure 2. Variance Inflation Factor (VIF) analysis identifying multicollinearity among indicators.

	Variable	VIF
8	Participation_Rate	82222.574071
9	Employment_Rate	72431.257150
10	Unemployment_Rate	451.423846
1	Immigrants	142.151604
7	Unaffordable_Housing	126.318393
6	Core_Housing_Need	73.069198
0	LIM_AT	60.175077
2	Visible_Minority	55.788292
11	Lone_Parent_Families	49.952634
3	Renters	40.311917
5	Bachelors_Degree	38.391348
4	No_Diploma	26.112923

Figure 3. Principal Component Analysis explained variance results.



	PC1	PC2	PC3
Visible_Minority	0.374051	0.063192	-0.194827
Immigrants	0.373515	0.107178	-0.222919
Lone_Parent_Families	0.363687	0.030531	-0.291463
Core_Housing_Need	0.359090	0.213565	0.033324
No_Diploma	0.329238	-0.072547	-0.334465
Unemployment_Rate	0.280388	-0.301435	0.240520
LIM_AT	0.270804	-0.094542	0.538964
Unaffordable_Housing	0.221315	0.120720	0.571136
Renters	0.193070	0.447541	0.184717
Bachelors_Degree	0.132083	0.474414	-0.062572
Participation_Rate	-0.206681	0.444393	0.042337
Employment_Rate	-0.229323	0.442145	-0.008071

Figure 4. Neighbourhood clustering based on the first two principal components.

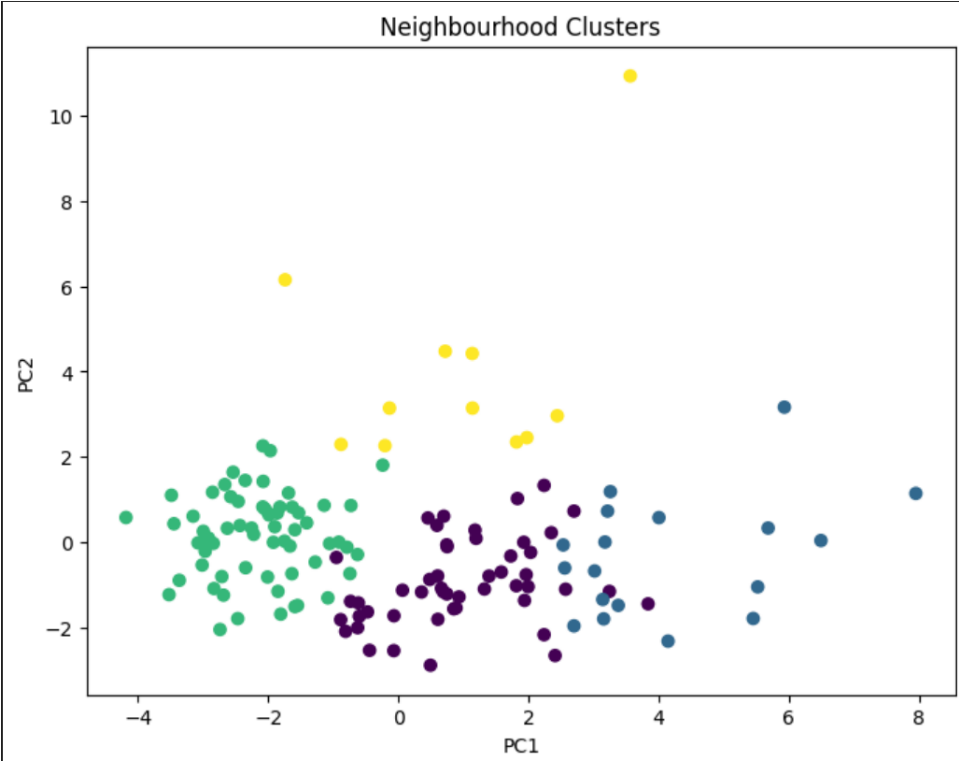


Figure 5. Average socioeconomic characteristics by cluster.

	LIM_AT	Immigrants	Visible_Minority	Renters	No_Diploma	Bachelors_Degree	Core_Housing_Need	Unaffordable
Cluster								
0	25.40	9681.43	10995.10	3564.90	2776.84	2951.94	1870.31	
1	23.03	20400.28	26368.06	4639.72	6312.22	5023.61	2999.72	
2	13.67	4687.50	3900.16	2391.53	1592.50	3086.77	980.56	
3	20.48	12162.27	11879.09	10847.73	2624.09	9837.27	3125.91	

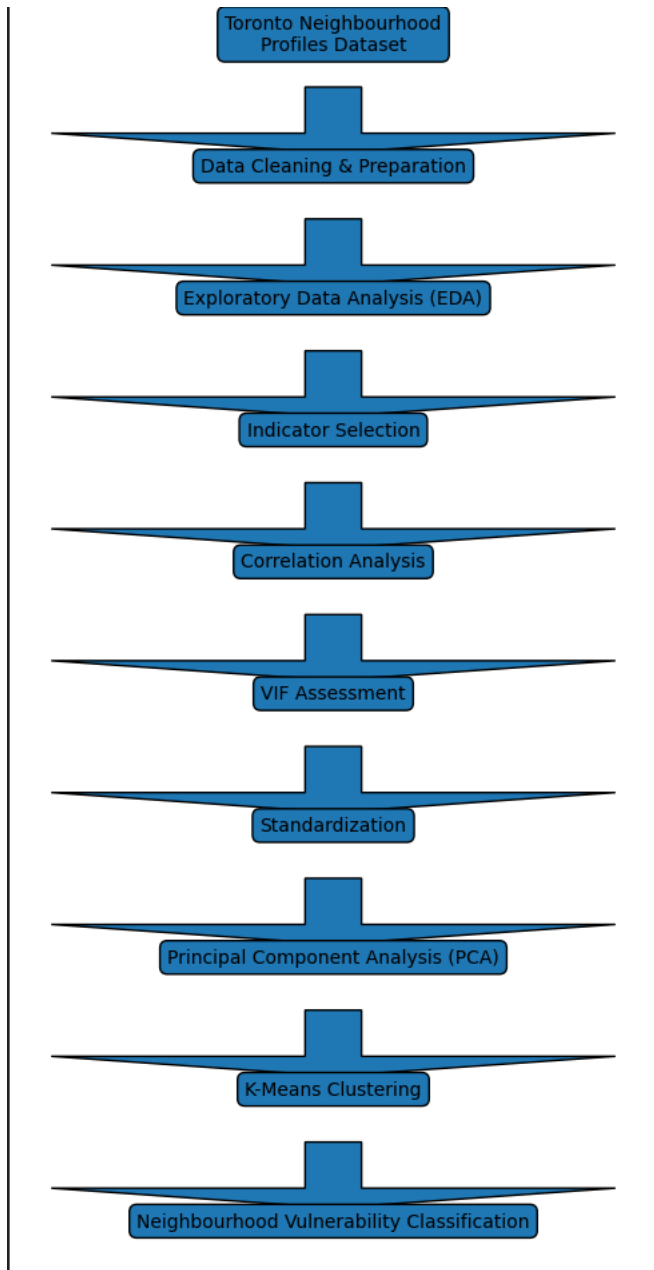
System Architecture and Sequence of operations

Figure 6. Pipeline Diagram

The analytical pipeline begins with the Toronto Neighbourhood Profiles dataset. Raw data have been imported into Python and cleaned through variable selection, formatting, and validation checks. Exploratory Data Analysis (EDA) is then conducted to examine distributions, identify potential outliers, and understand relationships among indicators.

Following EDA, correlation analysis and Variance Inflation Factor (VIF) assessment are performed to evaluate multicollinearity among socioeconomic variables. Because several indicators demonstrate strong correlations and high VIF values, the data are standardized and transformed using Principal Component Analysis (PCA).

The retained principal components are subsequently used as inputs to the K-Means clustering algorithm. Cluster assignments are then analyzed to identify neighbourhood groups with distinct socioeconomic characteristics and vulnerability profiles. Finally, cluster summaries and visualizations are generated to support interpretation and policy-oriented discussion.



Data Integration Strategy

The current phase of the project uses a single Toronto Neighbourhood Profiles dataset containing socioeconomic indicators for 140 neighbourhoods. Since all selected variables originate from the same source, no data merging was required during the exploratory analysis, dimensionality reduction, and clustering stages.

However, the next phase of the project will incorporate an additional Toronto Neighbourhood Profiles dataset from an earlier census period (2006) to support longitudinal analysis and predictive modelling. The 2006 and 2016 datasets will be integrated using common neighbourhood identifiers and neighbourhood names. Data integration will be performed using Pandas merge operations to align neighbourhood-level indicators across time periods.

The combined dataset will allow the analysis of socioeconomic changes over time, including trends in low-income prevalence, housing affordability, employment conditions, educational attainment, and demographic characteristics. This longitudinal structure will support Research Question 3 by providing the foundation for forecasting neighbourhood vulnerability and identifying areas experiencing increasing or decreasing socioeconomic risk.

Additional external datasets related to housing affordability, labour market conditions, or census indicators may also be incorporated in future phases if compatible neighbourhood-level identifiers are available. These datasets would be integrated using the same neighbourhood-based join strategy to enhance predictive modelling and improve the robustness of vulnerability assessments.

Conclusion

This interim report established the methodological foundation for the project. Data profiling identified significant multicollinearity among socioeconomic indicators, supporting the use of Principal Component Analysis (PCA) for dimensionality reduction. The combination of PCA and K-Means clustering provides an interpretable framework for identifying neighbourhood vulnerability patterns while satisfying the project's transparency requirements and the course requirement to avoid black-box methods.

The next phase of the project will focus on integrating historical Toronto Neighbourhood Profiles data from 2006 to support longitudinal analysis and predictive modelling. This will enable the investigation of changes in socioeconomic vulnerability over time and provide a foundation for forecasting neighbourhood-level risk patterns across Toronto.

GenAI declaration

Generative Artificial Intelligence (GenAI) tools- Microsoft Copilot, ChatGPT, Google Gemini, Cursor were used during this project as a supplementary learning, brainstorming and productivity aid. GenAI assistance was used to support code debugging, interpretation of statistical outputs, documentation drafting, report organization, and clarification of analytical concepts related to exploratory data analysis (EDA), Principal Component Analysis (PCA), and K-Means clustering.

GenAI was not used to generate or fabricate research findings, data, results, references, or conclusions. All data preparation, cleaning, transformation, statistical analysis, dimensionality reduction, clustering, visualization generation, and result validation were performed by the author using Python and Jupyter Notebooks. All code was executed and tested by myself, and all outputs were independently reviewed and verified before inclusion in the project.

I manually reviewed and edited all AI-assisted codes and ensured that methodological decisions, interpretations, and conclusions accurately reflected the analysis performed. Any suggestions provided by GenAI were evaluated against course materials, scholarly literature, and project requirements before implementation.

I confirm the ability to explain, justify, reproduce, and defend all aspects of the submitted work, including the analytical methodology, code implementation, statistical findings, and written report content.

References

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Git Hub Reference:

Microsoft. (n.d.). Introduction to GitHub. Microsoft Learn. (GitHub Structure & Reproducibility)